

Compactness implies discrete spectrum for arbitrary Kaplansky–Hilbert dynamical systems

Candidate full solution to Remark 3.8 of arXiv:2505.14132

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Status. Candidate full solution; likely valid, pending expert review. The result answers affirmatively the implication left open in Remark 3.8 of Haase–Kreidler [1]: for an arbitrary KH-dynamical system, order-density of the conditionally almost-periodic vectors implies discrete spectrum. In fact, every such vector already belongs to the discrete-spectrum part.

1 Source question

Let $(\mathbb{A}, S; E, T)$ be a KH-dynamical G -system: $\mathbb{A}$ is a Stone algebra, E is a Kaplansky–Hilbert module over $\mathbb{A}$, $S : G \rightarrow \text{Aut}(\mathbb{A})$, and each T_t is an S_t -covariant unitary. Write

$$E_{\text{ds}} = \text{ocl} \bigcup \{M \subseteq E : M \text{ is a finite-rank, } T\text{-invariant KH-submodule}\}.$$

A vector $x \in E$ is *conditionally almost periodic* if, for every $\varepsilon > 0$, there is a finite $F \subseteq E$ such that

$$\inf_{y \in F} |T_t x - y| \leq \varepsilon \mathbf{1} \quad (t \in G). \quad (1)$$

The source asks whether order-density of these vectors forces $E = E_{\text{ds}}$ [1, Remark 3.8]. The source proves this for KH-systems coming from measure-preserving extensions, using an ambient global L^2 -space, but explicitly leaves the abstract KH case open.

decreases to 0, where F ranges over the finite subsets of $L^\infty(X)$. Since the $L^1(Y)$ -norm is order-continuous [2, Sec.7.2], there is a finite set $F \subseteq L^\infty(X)$ with

$$\int_Y \sup_{t \in G} \inf_{g \in F} |T_t f - g|_Y \leq \varepsilon.$$

Now for $t \in G$ and $h \in F$ we certainly have

$$|T_t f|_Y^2 \leq |(T_t f | T_t f - h)_Y| + |(T_t f | h)_Y| \leq |T_t f - h|_Y + \sum_{g \in F} |(T_t f | g)_Y|,$$

and taking the infimum over h we arrive at

$$|T_t f|_Y^2 \leq \inf_{g \in F} |T_t f - g|_Y + \sum_{g \in F} |(T_t f | g)_Y| \leq \sup_{s \in G} \inf_{g \in F} |T_s f - g|_Y + \sum_{g \in F} |(T_t f | g)_Y|.$$

Integrating over Y we find, with $n := \#F$ being the cardinality of F ,

$$\|f\|_2^2 = \|T_t f\|_2^2 = \int_Y |T_t f|_Y^2 \leq \varepsilon + \sum_{g \in F} \|(T_t f | g)_Y\|_1 \leq \varepsilon + n^{\frac{1}{2}} \left(\sum_{g \in F} \|(T_t f | g)_Y\|_2^2 \right)^{\frac{1}{2}}$$

for each $t \in G$. Hence, it suffices to prove

$$\inf_{t \in G} \sum_{g \in F} \|(T_t f | g)_Y\|_2^2 = 0.$$

But this is true by [1, Prop.8.8], since $F \subseteq L^\infty(X)$. $\square$

As a corollary we obtain the announced characterization of extensions with discrete spectrum.

Corollary 3.7. *Let $X|Y$ be an extension of measure-preserving systems. Then the following assertions are equivalent.*

- (i) *The extension $X|Y$ has discrete spectrum.*
- (ii) *The extension $X|Y$ is compact, i.e., the space $\text{AP}_T(X|Y)$ is dense in $L^2(X)$.*
- (iii) *The set $\{f \in L^2(X|Y) \mid T_G f \text{ is totally order-bounded}\}$ is dense in $L^2(X)$.*
- (iv) *For every $f \in L^2(X)$ and every $\delta \in \mathbb{R}_{>0}$ there is a measurable set $E \in \Sigma_Y$ with $\mu_Y(E^c) \leq \delta$ such that $\mathbf{1}_E f \in \text{AP}_T(X|Y)$.*

Remark 3.8. By Lemma 3.1 we can reformulate assertion (ii) in Corollary 3.7 as

- (ii)' *The set $\text{AP}_T(X|Y)$ is order-dense in $L^2(X|Y)$.*

And (iii) can be reformulated as

- (iii)' *The set $\{f \in L^2(X|Y) \mid T_G f \text{ is totally order-bounded}\}$ is order-dense in $L^2(X|Y)$.*

Since also (i) can be reformulated entirely in terms of KH-modules, the equivalence of (i), (ii), and (iii) could actually be formulated for arbitrary KH-dynamical systems. However, whereas the implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) are still valid in this more general situation, we do not know whether this is also true for the implication (ii) $\Rightarrow$ (i). Our proof of Corollary 3.7 hinges on the Birkhoff–Alaoglu theorem, which refers to the global L^2 -dynamics, a feature not present in the abstract framework of KH-dynamical systems.

2 Proof intuition

Condition (1) supplies a finite-rank module whose orthogonal projection P almost contains the whole orbit of x . Conjugating P back along the dynamics produces finite-rank projections Q_t which all almost fix the *same* vector x . We take the order-closed cyclic convex hull of the Q_t inside the KH-module $\text{HS}(E)$. A lattice-valued version of the minimum-norm argument for Hilbert spaces gives a fixed point in this hull. It is an invariant Hilbert–Schmidt operator A_ε , and the hull construction preserves $|x - A_\varepsilon x| \leq \varepsilon \mathbf{1}$. The equivariant spectral theorem puts the range of every invariant Hilbert–Schmidt operator in E_{ds} . Letting $\varepsilon \downarrow 0$ proves $x \in E_{\text{ds}}$.

The cyclic (mix-closed) hull is the crucial feature: it lets one select the smaller of two $\mathbb{A}$ -valued norms on complementary idempotent pieces, which is what makes the minimum-norm element exist.

3 A cyclic convex fixed-point lemma

For a bounded subset C of a KH-module, call C *mix-closed* if every mixing $\sum_i p_i z_i$, with (p_i) a partition of unity by idempotents and $z_i \in C$, belongs to C .

Lemma 1 (cyclic convex fixed point). *Let $(\mathbb{A}, R; H, U)$ be a KH-dynamical G -system. Every nonempty bounded, order-closed, convex, mix-closed, U -invariant subset $C \subseteq H$ contains a fixed point of U .*

Proof. Set

$$h := \inf_{z \in C} |z|^2 \in \mathbb{A}_+.$$

The set of squared norms is downward directed. Indeed, for $u, v \in C$, the Stone property supplies an idempotent $p \in \mathbb{A}$ which selects the two order regions so that

$$p|u|^2 + p^c|v|^2 = |u|^2 \wedge |v|^2.$$

(Under $\mathbb{A} = C(\Omega)$, take the clopen closure of the region $|u|^2 < |v|^2$; on its boundary the functions agree.) Since C is mix-closed, $w = pu + p^c v$ belongs to C , and $|w|^2 = |u|^2 \wedge |v|^2$.

For every nonempty finite $K \subseteq C$, choose $z_K \in C$ with

$$r_K := |z_K|^2 = \bigwedge_{z \in K} |z|^2.$$

Ordered by inclusion, $r_K \downarrow h$. Convexity puts $(z_K + z_L)/2$ in C , so the parallelogram identity gives

$$\begin{aligned} |z_K - z_L|^2 &= 2r_K + 2r_L - 4 \left| \frac{z_K + z_L}{2} \right|^2 \\ &\leq 2(r_K - h) + 2(r_L - h). \end{aligned}$$

For $K, L \supseteq K_0$, the last expression is at most $4(r_{K_0} - h)$, which decreases to zero. Hence (z_K) is order-Cauchy. Order completeness and order-closedness give $z_K \rightarrow z_0 \in C$ in order, and order continuity of the modulus yields $|z_0|^2 = h$.

This minimizer is unique: if $|u|^2 = |v|^2 = h$, applying the same identity to their midpoint gives $|u - v|^2 \leq 0$. Finally, $U_t C = C$ and covariance imply

$$R_t h = \inf_{z \in C} R_t |z|^2 = \inf_{z \in C} |U_t z|^2 = h.$$

Thus $U_t z_0$ is another minimizer, and uniqueness gives $U_t z_0 = z_0$ for every $t \in G$. $\square$

4 Full answer

The implemented action on $\text{HS}(E)$ is

$$\mathcal{T}_t(B) = T_t B T_{t-1}^{-1}.$$

The Hilbert–Schmidt space is a KH-module and $\mathcal{T}$ is an S -covariant unitary action [2, Secs. 5–6]. Its fixed space is $\text{HS}_T(E)$, the invariant Hilbert–Schmidt operators.

Theorem 1. *For every KH-dynamical system $(\mathbb{A}, S; E, T)$,*

$$\text{AP}_T(E) \subseteq E_{\text{ds}}.$$

Consequently, the following are equivalent:

1. *E has discrete spectrum, i.e. $E = E_{\text{ds}}$;*
2. *$\text{AP}_T(E)$ is order-dense in E .*

In particular, the open implication (ii) $\Rightarrow$ (i) in Remark 3.8 of [1] is true for arbitrary KH-dynamical systems.

Proof. Fix $x \in \text{AP}_T(E)$ and $\varepsilon > 0$. Choose a finite set $F \subseteq E$ satisfying (1), put $m = |F|$, and let

$$M = \text{ocls span}_{\mathbb{A}} F.$$

This is a finite-rank KH-submodule of rank at most m [2, Sec. 2.5]. Let P be its orthogonal projection. The Pythagorean identity gives, for $z \in E$ and $y \in M$,

$$|(I - P)z|^2 \leq |z - y|^2.$$

Taking the infimum over $y \in F$ in (1) therefore yields

$$|(I - P)T_t x| \leq \varepsilon \mathbf{1} \quad (t \in G). \quad (2)$$

For $t \in G$ define the $\mathbb{A}$ -linear orthogonal projection

$$Q_t = T_{t-1} P T_t = \mathcal{T}_{t-1}(P).$$

Since P has rank at most m , $P \in \text{HS}(E)$ and $|P|_{\text{HS}}^2 \leq m \mathbf{1}$. Covariant unitarity gives the same bound for every Q_t . Moreover, by (2),

$$|x - Q_t x| = S_{t-1} |(I - P)T_t x| \leq \varepsilon \mathbf{1}. \quad (3)$$

Inside the lattice ball $\{B \in \text{HS}(E) : |B|_{\text{HS}} \leq \sqrt{m} \mathbf{1}\}$, let C be the smallest order-closed, convex, mix-closed set containing the orbit $\{Q_t : t \in G\}$. Equivalently, intersect all such subsets of the ball. The family is nonempty because the ball itself is admissible. The implemented action permutes the orbit and preserves each of the defining properties, so $\mathcal{T}_t C = C$.

Lemma 1 supplies $A_\varepsilon \in C$ fixed by $\mathcal{T}$, hence $A_\varepsilon \in \text{HS}_T(E)$. The constraint set

$$D_\varepsilon = \{B \in \text{HS}(E) : |x - Bx| \leq \varepsilon \mathbf{1}\}$$

is order-closed, convex, and mix-closed. For order-closedness, use the bounded evaluation map

$$|Bx| \leq |B|_{\text{HS}} |x|.$$

By (3), D_ε contains every Q_t ; hence $C \subseteq D_\varepsilon$ and

$$|x - A_\varepsilon x| \leq \varepsilon \mathbf{1}. \quad (4)$$

The KH decomposition theorem identifies

$$E_{\text{ds}} = \text{ocl} \sum \{\text{ran } B : B \in \text{HS}_T(E)\} \quad [2, \text{Theorem 6.9}].$$

Thus $A_\varepsilon x \in E_{\text{ds}}$. Apply (4) with $\varepsilon = 1/n$. Then $A_{1/n}x \rightarrow x$ in norm (and hence in order), so the order-closedness of E_{ds} gives $x \in E_{\text{ds}}$. This proves $\text{AP}_T(E) \subseteq E_{\text{ds}}$.

If $\text{AP}_T(E)$ is order-dense, the inclusion forces $E_{\text{ds}} = E$. Conversely, every vector in an invariant finite-rank KH-submodule has a uniformly totally order-bounded orbit [1, Proposition 2.5]; the definition of E_{ds} then gives order-density of $\text{AP}_T(E)$ whenever $E = E_{\text{ds}}$. $\square$

5 Verification, scope, and novelty

- The proof uses no global L^2 realization, invariant measure, group amenability, or topology on G .
- The fixed-point lemma was checked directly from order completeness, mixing, convexity, and the parallelogram identity. The critical evaluation map is order-continuous because its lattice norm is dominated by $|B|_{\text{HS}}|x|$.
- A tempting rank-one convexification can collapse to zero when supports move under S . The conjugate-projection construction avoids this: every operator in the initial orbit approximately fixes the same x , and that closed cyclic convex constraint survives the fixed-point step.
- The conclusion is stronger than the requested density implication: each conditionally almost-periodic vector lies in E_{ds} .
- Bounded novelty searches on 13 August 2026 covered the run indexes, the revised arXiv source, and web/arXiv phrases combining “KH-dynamical system”, “conditionally almost periodic”, “order-dense”, “compact/discrete spectrum”, and “cyclic convex fixed point”. No later answer to this exact abstract implication was found. The source revision of 9 February 2026 still states it as open. Novelty confidence is moderate: the cyclic convex fixed-point lemma may be known in Banach–Kantorovich language even though this application was not located.

Human-review recommendation. Prioritize the three interfaces in the proof: (1) the least lattice-norm construction in Lemma 1; (2) invariance of the smallest bounded order-closed convex mix-closed hull under the semilinear implemented action; and (3) the use of the Edeko–Haase–Kreidler characterization of E_{ds} by ranges of invariant Hilbert–Schmidt operators. No other unproved dependency remains.

References

- [1] M. Haase and H. Kreidler, *Precompactness notions in Kaplansky–Hilbert modules and extensions with discrete spectrum*, arXiv:2505.14132v3, 2026.
- [2] N. Edeko, M. Haase, and H. Kreidler, *A decomposition theorem for unitary group representations on Kaplansky–Hilbert modules and the Furstenberg–Zimmer structure theorem*, Analysis Mathematica 50 (2024), 377–454; arXiv:2104.04865.